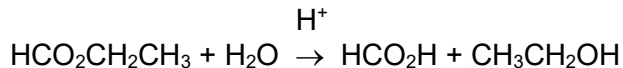


- 12 Ethyl formate undergoes a slow acid-catalysed hydrolysis in water.



The rate law is found to be

$$\text{rate} = k[\text{HCO}_2\text{CH}_2\text{CH}_3][\text{H}^+]$$

When 0.1 mol dm^{-3} of HCl is reacted with 0.4 mol dm^{-3} of ethyl formate, the half-life was found to be 62 min.

Another reaction was carried out with 0.3 mol dm^{-3} of HCl and 0.4 mol dm^{-3} of ethyl formate.

How long does it take for the concentration of ethyl formate to fall to $0.050 \text{ mol dm}^{-3}$?

- A** 31 min **B** 62 min **C** 93 min **D** 124 min